Assignment 5 - Sequencer and Rendering

Introduction

Modern Tech- A5

5

In this module, you will explore the basics of three key modules within Unreal for creating animated sequences or cinematics.

Sequencer

Sequencer is essentially the Camera Package Plugin for UE. It includes a Timeline for animations, editing and cutting between cameras to create cut scenes and in-game camera moves. While there are a lot of advanced features here to explore, we will just be looking at the basics.

Take Recorder

Take Recorder is a plugin that allows you to record live gameplay to a sequence track, which you can then play back using sequencer. You could also use it for player "Ghosts". It is especially helpful when making cutscenes as you can capture natural looking gameplay and animation without having to hand place every animation clip and effect.

Render Queue

The Render Queue is the updated UE renderer. It can produce high quality renders of multiple cameras, and allows for batching, customization of the output, and ties in with the Sequencer to create rendered cut-scenes from game play elements.

Gettings Started / FAQ

Note: You should do this in your own solo test project until you get a handle on the topics.

I was working from the Side Scroller 3D Person Template and recommend that as a starting point.

These are great tools for rendering out final footage for cutscenes however, so you'll want to do this (outside of this assignment) to support a trailer, highlighting art assets, and creating portfolio pieces.

Task 1-3 Sequencer

Part 1: Sequencer

Task 1: Set up Level Sequence and create a Cine Camera.

- Frame your subject with a Cine Camera by moving the camera into position, and setting the focus distance so the subject is clear in your shot.

Task 2: Animate the Cine Camera movement to create an interesting animated sequence.

- Set at least 3 keyframes on the Transform track of your Cine Camera.
- Include at least 2 different views of the world using the same lens *(e.g close up zooms in to a mid character shot, or start at the character, and pan up to reveal the environment etc.)*
- Keyframe at least one of the settings on the camera to change in a way that supports the shot.
- Keyframe at least one of the settings on the background or actor to support the shot.
- Your final sequence should be ~5-20 seconds long.

Task 3: Set up at least 3 distinct shots of your recorded Player (from the Take Recorder), using the Cut Track and 3 different Cine Cameras.

- You should do one Close Up (long lens), one Wide environment shot (short lens), and one Medium shot (character is the focus).
- Create a camera Cut Track that cuts between the three different cameras.

Deliverables:

☐ ***Task 1 Deliverable:*** *A screenshot showing your Sequencer Timeline and the static shot you created.*

☐ ***Task 2 Deliverable:*** *A short video of the shot you created. Alternatively, you could submit a screenshot of your timeline, and 3-5 screenshots showing the "key frames" of your shot.*

☐ ***Task 3 Deliverable:*** *Either the exported video file of your camera cut track playing, or a screenshot showing your Sequencer Timeline demonstrating it was completed, and a screenshot of each of your distinct shots.*

** Note: If you want to render out your frames, make sure you combine them back into video files, or just screen record within UE. Alternatively, render out a PNG sequence and hand in 3 frames that show your 3 distinct shorts. Don't hand in your full raw PNG sequence for these 3 tasks.*

Task 4 Take Recorder

Part 2: Take Recorder

Task 4: Open the Take Recorder plugin and record the Player moving through the level.

- 10-15 seconds should be sufficient.
- Plug the recorded Shot into the sequencer to play back the Player as an animation.
- There is a Player setting that will automatically capture the active player.
- You'll have to run the scene first, then activate Take Recorder.

☐ ***Task 4 Deliverable:*** *A short video showing your Sequencer Timeline and the Player animation you recorded. Alternatively, you could submit a screenshot of your timeline, and 3-5 screenshots showing the "key frames" of your player being animated.*

Task 5 & 6 Render Queue

Part 3: Render Queue

Task 5: Render out one of your sequences as a sequence of PNG frames using the Render Queue.

Task 6: Create a single video that includes your Task 2 Camera animation, and Task 3 shots (Close, Mid, Far). This should be edited together into a single video file.

- Note: You could either render out all 4 sequencers separately and combine them in 3rd party video editing tools such as DaVinci Resolve, or you could do your "edits" in the Camera Cuts track and export that.
- While it is possible to render video out of the Render Queue directly, it likely won't work out of the box and you'll have to hook it up to the command line FFMPEG encoder. I had issues getting this setup, so I'll just say do so at your own risk.

☐ ***Task 5 Deliverable: Either the final video (as an MP4 h264 video file) created from the frames, or a selection of 5 of the frames themselves showing progression*** *(e.g. if you had 40 frames, include frame 0, 10, 20, 30, 40, rather then 0,1,2,3,..). Don't include the full raw PNG sequence.*

☐ ***Task 6 Deliverable: The final video clip including all 4 shots, exported as a single MP4 h264 file.***

Note: MP4 is the file type or container, while h264 is the Codec type used to encode it. Other types are not guaranteed to work when Imarking.

Grading

Grading Details - Rubric

Rubric

Tasks:

There are 6 Tasks to complete. Each are worth 3 Points.

Your total assignment grade will be [_/18]

Each Task is graded either:

3/3 - Successfully Completed the task. Progress is clearly demonstrated and presented in a professional manner.

2/3 - Task is mostly completed, but may be missing elements, contain errors or is presented unprofessionally.

1/3 - Task is attempted, with a demonstrable effort made, but is not finished correctly or has multiple errors.

0/3 - Not Completed

Assignment 5 will be worth 15% of your grade in this course.

Grading Details - Basic Video Content Guidelines

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Basic Video Submission Requirements:

Format Guidelines:
For this type of deliverable, the video can be raw screen recording and doesn't need to have editing, voice over or subtitles. The video is just to serve as an easy to reference visual showing what you did in the modification task.

Screen recordings must be submitted as MP4 (H.264) files.

Recommended tools: OBS Studio or Windows Snipping Tool for screen capture.

DaVinci Resolve is a full video editing suite for more advanced manipulation, but you can use any software you like to combine frame sequences into video.

Resolution should be 720p or 1080p at 30 FPS.

Use a reasonable compression ratio (should be 50-300 MB). Learn won't accept files over 1GB.

The MP4 file should have a reasonable name (e.g GD2_A1_DylanFries.MP4). Videos should NOT be placed in a ZIP (or other type) of archive file.

Submissions in other formats may not be graded.

LINKS (e.g. to Youtube) **WILL NOT BE ACCEPTED.**